

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Pharmacy SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Pharmacy Branch : Pharmacy

Semester : SEMESTER - 4

Subject Code & Name: BP403T - PHYSICAL PHARMACEUTICS II

Time : 3 Hours]

[Total Marks : 75

Instructions to the Students:

- 1.All questions are compulsory.
- 2.Draw diagrams / figures wherever necessary
- 3.Figures to right indicate full marks

Q.1 Objective Type Questions (Answer all the questions) (10 x 2) = 20

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|--|---|
| A) Write Pseudo first order reaction with example. | 2 |
| B) Enlist Physical factors influencing the chemical degradation of pharmaceutical product. | 2 |
| C) What is rheology? | 2 |
| D) Define Emulsions. | 2 |
| E) Define phase Inversions. | 2 |
| F) comment on deformation of solids. | 2 |
| G) Differentiate between lyophilic & lyophobic colloids. | 2 |
| H) Write in short about Zeta Potential. | 2 |
| I) Enlist various applications of chemical kinetics in pharmacy. | 2 |
| J) What is gold number give example. | 2 |

Q.2 Long Answers (Answer 2 out of 3) (2 x 10) = 20

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|---|----|
| A) Define Zero order reaction with suitable examples. Deduce an equation for the determination of rate constant, half life & shelf life for zero order reaction kinetics. | 10 |
| B) Discuss the optical & electrical properties of colloids. | 10 |
| C) With the help of neat diagram explain the working principle of Cup & bob and Cone & plate viscometer with its advantages & disadvantages. | 10 |

Q.3 Short Answers (Answer 7 out of 9) (7 x 5) = 35

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|---|---|
| A) Explain the factors influencing the rate of a reaction. | 5 |
| B) Write an note on accelerated stability study. | 5 |
| C) Define thixotropy & write importance of thixotropy in formulation. | 5 |
| D) Write a note on identification tests of emulsions. | 5 |
| E) Define & Classify suspension with examples. | 5 |
| F) Explain in brief flow properties of powder. | 5 |
| G) Explain electrical double layer theory with well labelled sketch | 5 |
| H) Explain the optical, kinetic & Property of colloidal dispersion | 5 |
| I) Enumerate methods to determine the particle size. Explain any two methods. | 5 |

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